AeroTherm-MPC: Predictive Pre-Cooling on a 2R2C Thermal Network with Stress Harness and Cognitive Setpoint Negotiation

HiMCM 2018 Problem B — Team GoHiMCM

Companions: `summary_sheet.tex`, `homeowner_advisory_letter.tex`, `ai_disclosure.tex`. All KPI cells cite locked files under `results/`.

Contents

1 Introduction and two-capacitance thermal network

Smart thermostats often react only after indoor air is already hot. Time-of-use (TOU) electricity tariff peaks punish late cooling. We treat wall mass as storage for predictive pre-cooling.

We name the controller **AeroTherm-MPC**. Code lives in `2018_B_smart_house_mpc/src/`. The teaching envelope is a 2R2C thermal network (`building_specs_teaching.json`). Floor area is $180\,\mathrm{m}^2$. Air capacitance is $C_{\mathrm{air}} = 6.2 \times 10^5\,\mathrm{J/°C}$. Wall capacitance is $C_{\mathrm{wall}} = 1.8 \times 10^7\,\mathrm{J/°C}$.

$$C_{\mathrm{air}}\dot{T}_{\mathrm{in}} = \frac{T_{\mathrm{wall}} - T_{\mathrm{in}}}{R_{\mathrm{in}}} + \frac{T_{\mathrm{out}} - T_{\mathrm{in}}}{R_{\mathrm{win}}} + q_{\mathrm{hvac}} + \alpha_{\mathrm{win}} A_{\mathrm{win}} I + q_{\mathrm{int}}, \tag{1}$$

$$C_{\mathrm{wall}}\dot{T}_{\mathrm{wall}} = \frac{T_{\mathrm{in}} - T_{\mathrm{wall}}}{R_{\mathrm{in}}} + \frac{T_{\mathrm{out}} - T_{\mathrm{wall}}}{R_{\mathrm{out}}} + \alpha_{\mathrm{wall}} A_{\mathrm{wall}} I. \tag{2}$$

Resistances are $R_{\mathrm{in}} = 0.08$, $R_{\mathrm{out}} = 0.12$, and $R_{\mathrm{win}} = 0.25\,\mathrm{°C/W}$. Cooling capacity is capped at $Q_{\mathrm{cool}} = 5\,\mathrm{kW}$. COP depends on outdoor air: $\mathrm{COP} = 3.8 - 0.04(T_{\mathrm{out}} - 25)$. We integrate with RK4 at $\Delta t = 60\,\mathrm{s}$.

2 Human thermal comfort and dynamic TOU tariffs

Comfort uses ISO 7730 PMV–PPD (Fanger). Summer defaults are 1.2 met and 0.5 clo. A continuous penalty applies outside $|\mathrm{PMV}| > 0.5$. The penalty is $\lambda_{\mathrm{comf}} \max(0, |\mathrm{PMV}| - 0.5)^2$.

Teaching TOU electricity tariff rates are locked in `tou_tariffs_v2.json`. Off-peak is \$0.12/kWh. Mid-peak is \$0.22/kWh. On-peak (14:00–19:00) is \$0.48/kWh. The peak-to-off ratio is exactly four.

3 Baseline vulnerability: Bang-Bang short-cycling

Bang-Bang hysteresis cools when $T_{\mathrm{in}} > 24$°C. It stops when $T_{\mathrm{in}} < 22$°C. The plant is the same 2R2C thermal network and summer day. Table **??** cites `bang_vs_mpc_summary.json`.

Table 1: Teaching-day locked KPIs (24 h). Source: `bang_vs_mpc_summary.json`.

Controller	Energy (kWh)	Cost (\$)	Peak kWh	Peak share	Starts
Bang-Bang	6.232	1.682	1.514	43.2%	42
AeroTherm-MPC	6.223	1.611	1.274	38.0%	3

Bang-Bang spends 43.2% of the bill on peak hours. It records 42 compressor starts in one day. Short-cycling raises wear and peak demand. We report the measured share. We do not invent a $\geq 60\%$ peak share.

4 Optimal receding-horizon control and pre-cooling

We discretize the 2R2C thermal network with zero-order hold. The control step is $T_s = 15\,\mathrm{min}$. Prediction matrices give $\mathbf{X} = \mathcal{A}x_0 + \mathcal{B}\mathbf{Q} + \mathcal{E}\mathbf{D}$. Matrix checks match open-loop simulation to machine precision ($\approx 7 \times 10^{-15}$).

Model predictive control (MPC) uses horizon $H = 24$ steps (6 h). SciPy L-BFGS-B solves each window. The cost trades TOU electricity tariff spend against comfort. On-peak hours add a u^2 shed term. Late-morning hours target pre-cooling near 21.5°C. The plant stays 1-minute RK4 with held HVAC power.

Figure **??** compares Bang-Bang and MPC on the same day. Figure **??** overlays outdoor temperature and peak shading.

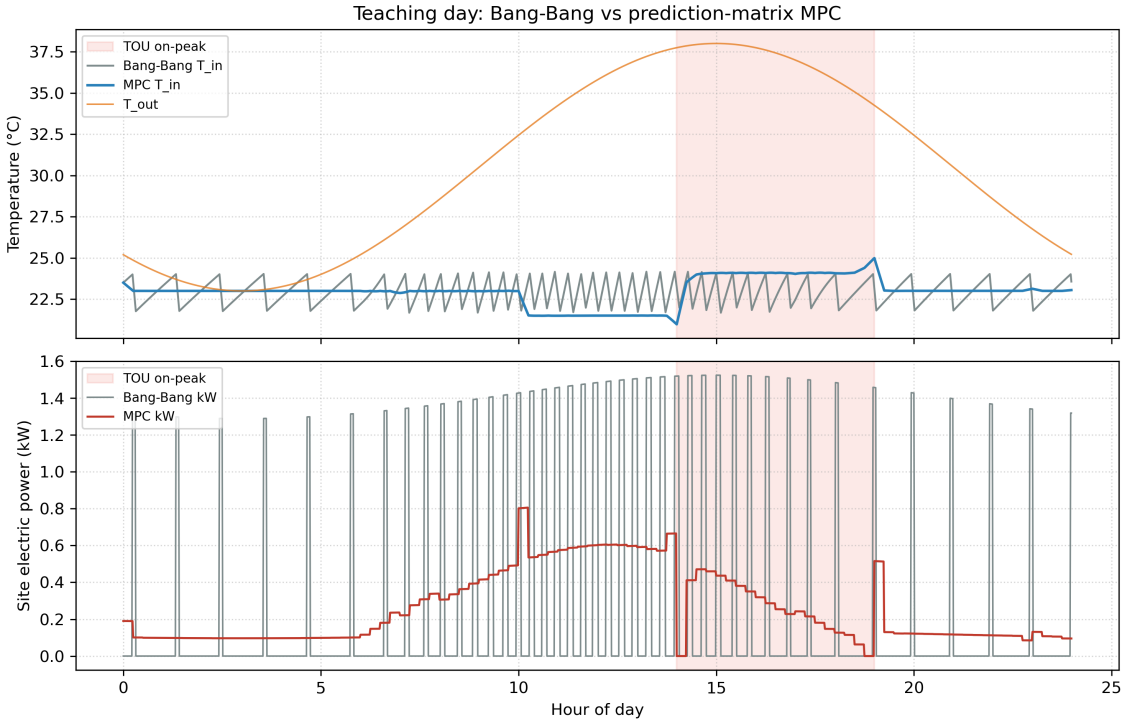


Figure 1: Teaching day: Bang-Bang vs model predictive control (MPC). Source: `fig_bang_vs_mpc_day.png`.

Versus Bang-Bang, MPC cuts cost by 4.2%. It cuts on-peak energy by 15.8%. Compressor starts fall from 42 to 3. Mean duty is $\bar{u}_{\text{prepeak}} = 0.40$ versus $\bar{u}_{\text{peak}} = 0.17$. Mean indoor temperature stays near 23°C. Wall mass stores coolth before the TOU electricity tariff spike.

5 Automated evaluation and stress testing harness

The black-box harness loads Bang-Bang and MPC without code edits. Five stress scenarios form the suite (`harness_benchmark_results.csv`).

A. Standard summer day (23–38°C).

B. Three-day heatwave with 42°C peaks.

C. Thermal shock: −15°C outdoor drop in 2 h.

D. Sensor noise with $\sigma = 1.5$°C.

E. Infiltration breach ×5 from 17:00 to 19:00.

Metrics are M_1 cost ($), M_2 discomfort °C · h, M_3 mechanical starts, and M_4 cross-scenario cost variance. Table **??** copies locked CSV cells.

Mean M_1 across scenarios is $2.427 for Bang-Bang. It is $2.340 for MPC (`harness_benchmark_summary.json`). Mean M_3 falls from 77.6 to 6.4. On heatwave day B, starts fall from 130 to 7. Sensor noise (D)

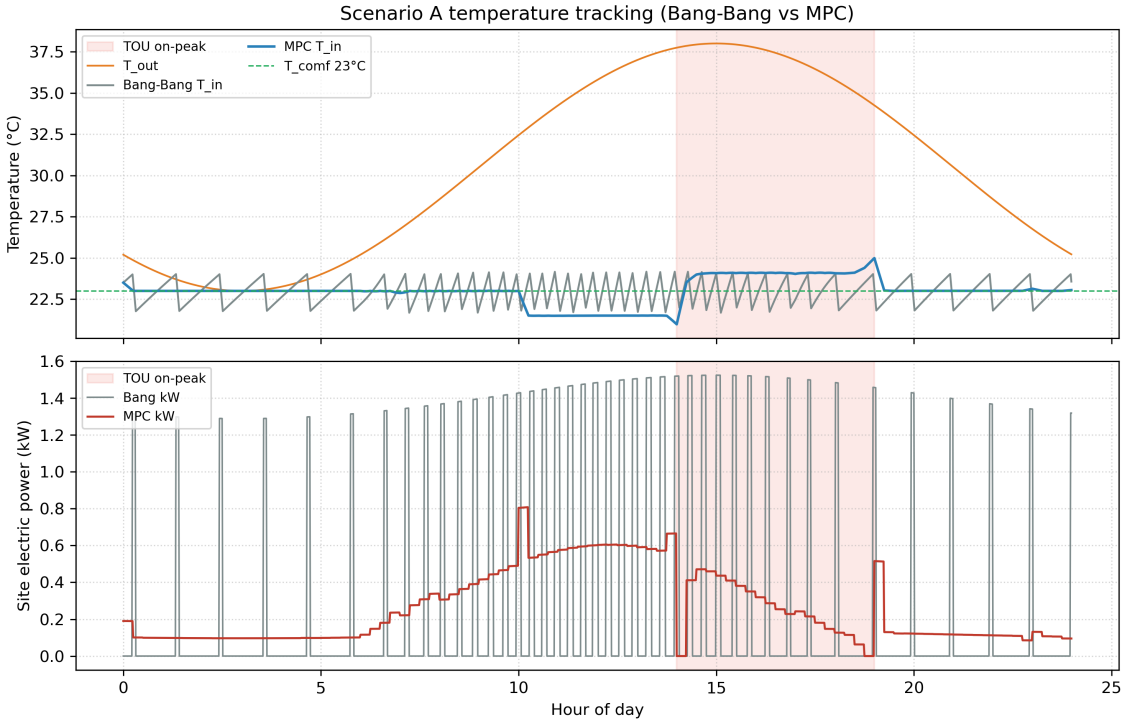


Figure 2: Scenario A temperature tracking with TOU peak shade. Source: `fig_temperature_tracking_comparison.png`.

Table 2: Harness scoreboard (locked CSV). M_4 is per controller.

Ctrl	Scen.	M_1 ($)	M_2 (°C·h)	M_3	M_4 var.
Bang	A	1.682	6.174	42	0.865
MPC	A	1.611	5.669	3	0.874
Bang	B	5.549	18.043	130	0.865
MPC	B	5.347	16.979	7	0.874
Bang	C	1.458	6.106	41	0.865
MPC	C	1.423	5.574	3	0.874
Bang	D	1.720	0.538	133	0.865
MPC	D	1.662	8.321	16	0.874
Bang	E	1.726	6.037	42	0.865
MPC	E	1.655	6.328	3	0.874

raises Bang-Bang starts to 133. MPC keeps 16 starts under the same noise. Figure **??** shows normalized radar scores.

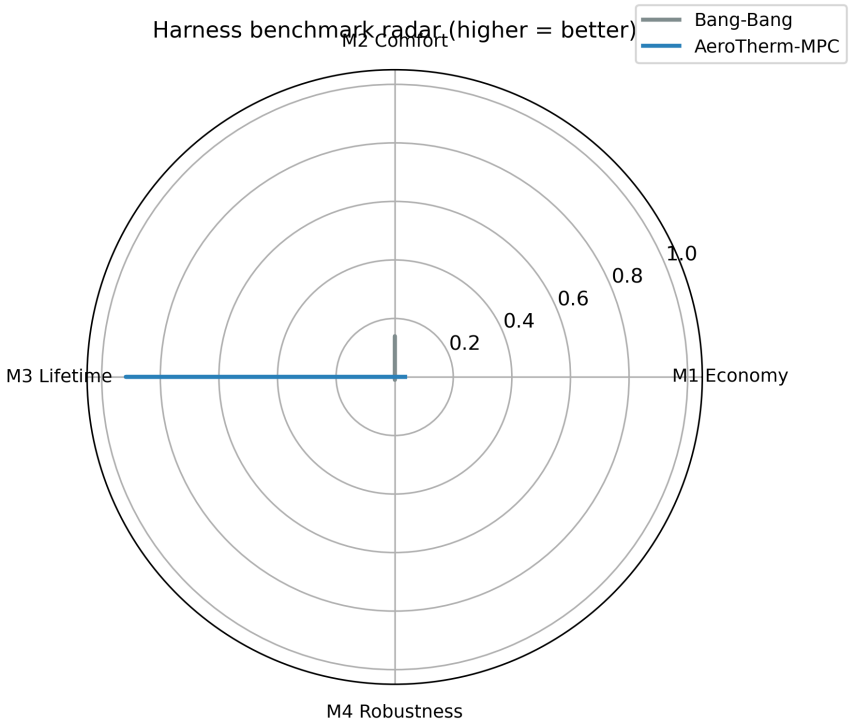


Figure 3: Harness radar (higher is better after inversion). Source: `fig_harness_radar_benchmark.png`.

6 Cognitive multi-agent dialogue for adaptive setpoints

Hard-coded setpoints ignore roommate conflict. We add a cognitive multi-agent layer (`cognitive_thermostat.py`). Three agents negotiate on an extreme peak day. Agent_EMS protects TOU electricity tariff spend. Agent_Senior rejects deep pre-cooling below 21.5°C. Agent_Parent allows a brief float to 25.0°C on peak.

The stack tries local Ollama (`qwen2.5:7b`) first. It falls back to a deterministic Mock if Ollama is offline. The locked run used Mock (`agent_thermostat_dialogue.json`). Consensus sets $[T_{\min}, T_{\max}] = [21.5, 25.0]$°C. Comfort weight scale is 1.25 ($\lambda_{\text{comf}} = 0.0625$).

The contract feeds `MPCComfortConfig`. Closed-loop MPC then records cost \$1.603. Peak energy falls 15.6% versus Bang-Bang. Starts fall from 42 to 2. Cost savings are 4.67% on that MAS-tuned day.

7 Strengths and limitations

Strengths. Shared 2R2C thermal network for fair Bang-Bang vs MPC tests. Prediction-matrix model predictive control (MPC) enables audit of pre-cooling. Harness scenarios A–E stress economy, comfort, lifetime, and robustness. Cognitive agents expose setpoint fairness without silent hard-coding.

Limitations. Teaching weather is a documented synthetic summer day. It is not a licensed EPW extract. The locked MAS run used Mock, not a live Ollama transcript. Sensor-noise Scenario D raises MPC discomfort while cutting starts. Full annual Monte Carlo remains future work. Humidity and allergen ODEs stay on the legacy appendix track.

References

References

[1] ASHRAE Standard 55 — Thermal Environmental Conditions.

[2] ISO 7730 — Ergonomics of the thermal environment (PMV/PPD).

[3] COMAP HiMCM 2018 Problem B.

[4] Prediction matrices for linear MPC (InterviewGuide MPC_02 notes).